

Andrew Xiao

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EDUCATION

Rutgers University – New Brunswick

Sep 2024 – May 2028

Bachelor of Arts in Computer Science, Minor in Economics

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Architecture, Discrete Math I & II, Linear Algebra

EXPERIENCE

Software Engineering Intern

Jun 2025 – Aug 2025

Mad Realities

Manhattan, NY

- Developed a RAG-based SMS chatbot integrating Community webhooks, Cloudflare, and Gemini, to **3,000+** users and boosting fan interaction by **30%**.
- Shipped an analytics dashboard (JS, SQL, Palantir Foundry) reducing report generation by 90% with automated data refresh pipelines instead of manual data reporting
- Optimized a Palantir Foundry analytics dashboard by implementing asynchronous time-series rendering and lazy loading, reducing load time by **80%** (5s → 1s)
- Led development of *The Love Channel* website (React, TypeScript, Cloudflare Workers, Firebase) reaching **2,000+** unique visits.

PROJECTS

ShuttleRank | *React Native*

Mar 2026 – Current

- Built a cross-platform rating app (web + iOS/Android) using React Native and a shared PostgreSQL database backed by Redis, enabling consistent skill tracking across devices.
- Designed a Glicko-2 based rating engine extended with a custom score-differential factor using tanh normalization, tracking per-player uncertainty and volatility across matches.

RU My Valentine | *FastAPI, Pydantic, SQLAlchemy, PostgreSQL, AWS EC2*

Oct 2025 – Feb 2026

- Designed a weighted vector compatibility algorithm that separates categorical constraints (gender, orientation) as hard pre-filters before computing preference similarity scores across quiz responses.
- Implemented OAuth, JWT authentication, and rate-limiting to prevent fraudulent quiz submissions and spam signups, securing a platform that launched to **2,300+** Rutgers students.

TFTMonitor | *FastAPI, Pydantic, psutil, Websocket*

Feb 2026 – Mar 2026

- Built a backend service enforcing per-user playtime limits by combining psutil process monitoring with LCU WebSocket event listening, detecting game-end within 500ms of the triggering event.
- Resolved race conditions in end-of-game detection by disambiguating between multiple concurrent LCU events, ensuring accurate session termination across edge cases.

LEADERSHIP

Club Manager

Oct 2022 – Jul 2024

New Jersey Badminton Club

Montville, NJ

- Implemented a membership system that increased revenue flow by 20% and boosted membership growth and retention by 15%.
- Organized a 200+ player tournament, ensuring smooth court allocation and efficient time management.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, HTML, CSS

Frameworks/Libraries: React, Next.js, FastAPI, SQLAlchemy, Pydantic, Tailwind CSS

Developer Tools: Git, Github, Node.js, psutil, Websocket, Palantir Foundry, Cloudflare Workers/D1, Firebase, Supabase, AWS EC2, Docker

Activities: HackRU, Collegiate Badminton Team (Division 1 tournaments)